# MatchCityLoc: Aerial Localization in LoD City Models via Surface Normal Matching

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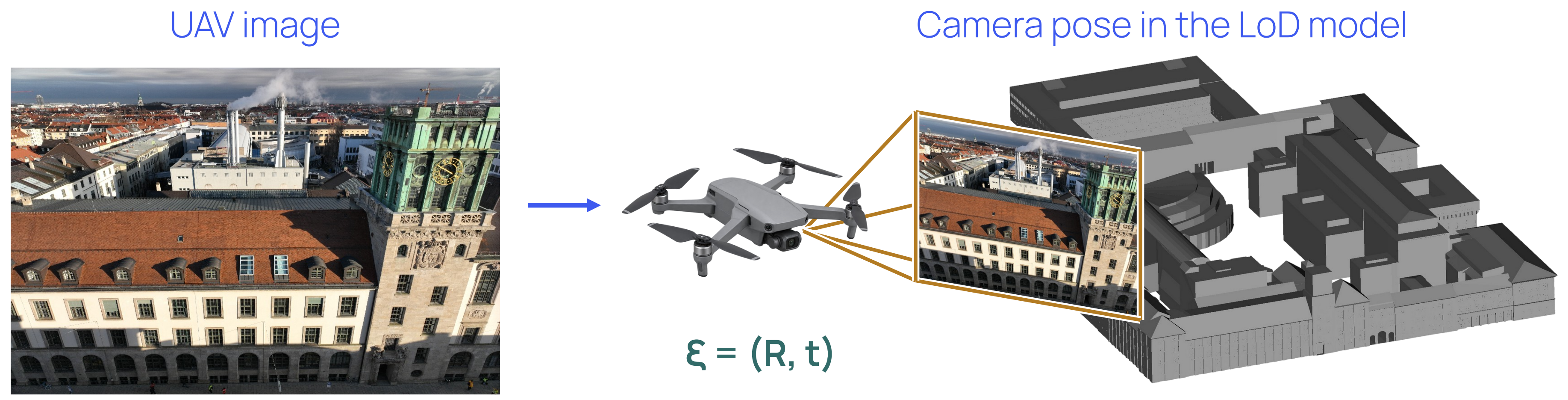
WACV JAN 4-8 2027 Orlando, Florida

## Goal & Motivation

Goal: refine a UAV camera's 6-DoF pose in a textureless city model.


UAV image
Camera pose in the LoD model
ξ = (R, t)

Known intrinsics + position-and-orientation prior

## Potential Applications

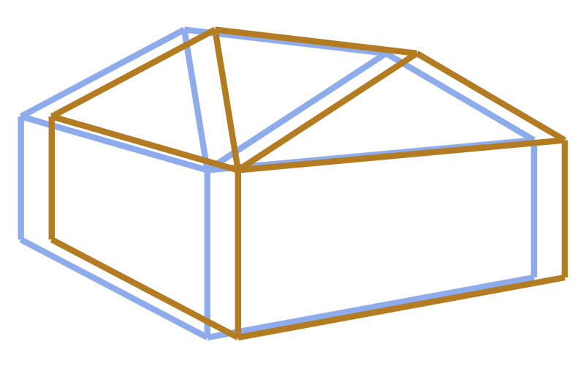
Urban navigation

Building inspection

Urban mapping

## Previous Work

**LoD-Loc v1 [1]**



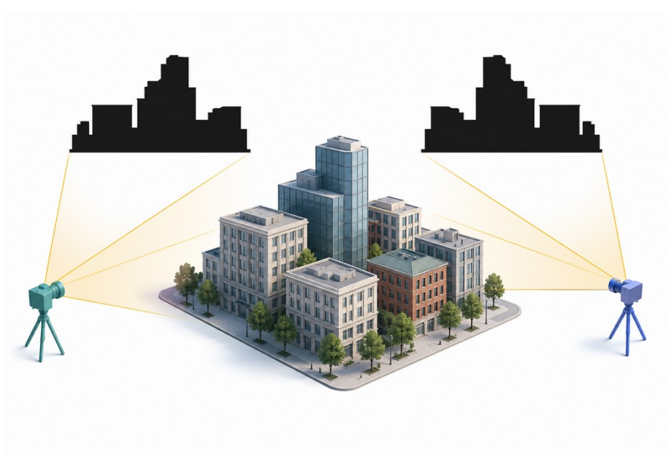
Predicted / projected edges

**Wireframe alignment**

Align predicted edges with projected 3D lines.

Limit: edge quality; dataset-specific training.

**LoD-Loc v2/v3 [2, 9]**
**SemCityLoc [3]**

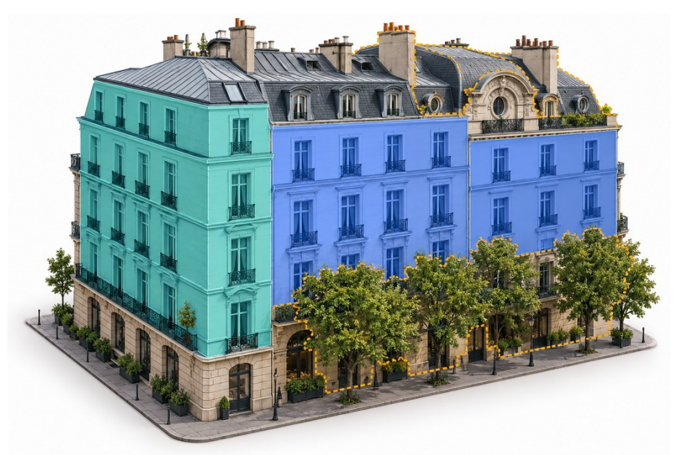


**Silhouette alignment**

Score pose hypotheses by mask/surface overlap.

Limit: ambiguous or inaccurate masks.

**PlanarCityLoc [4]**



**3D plane alignment**

Align image-derived planes with model planes.

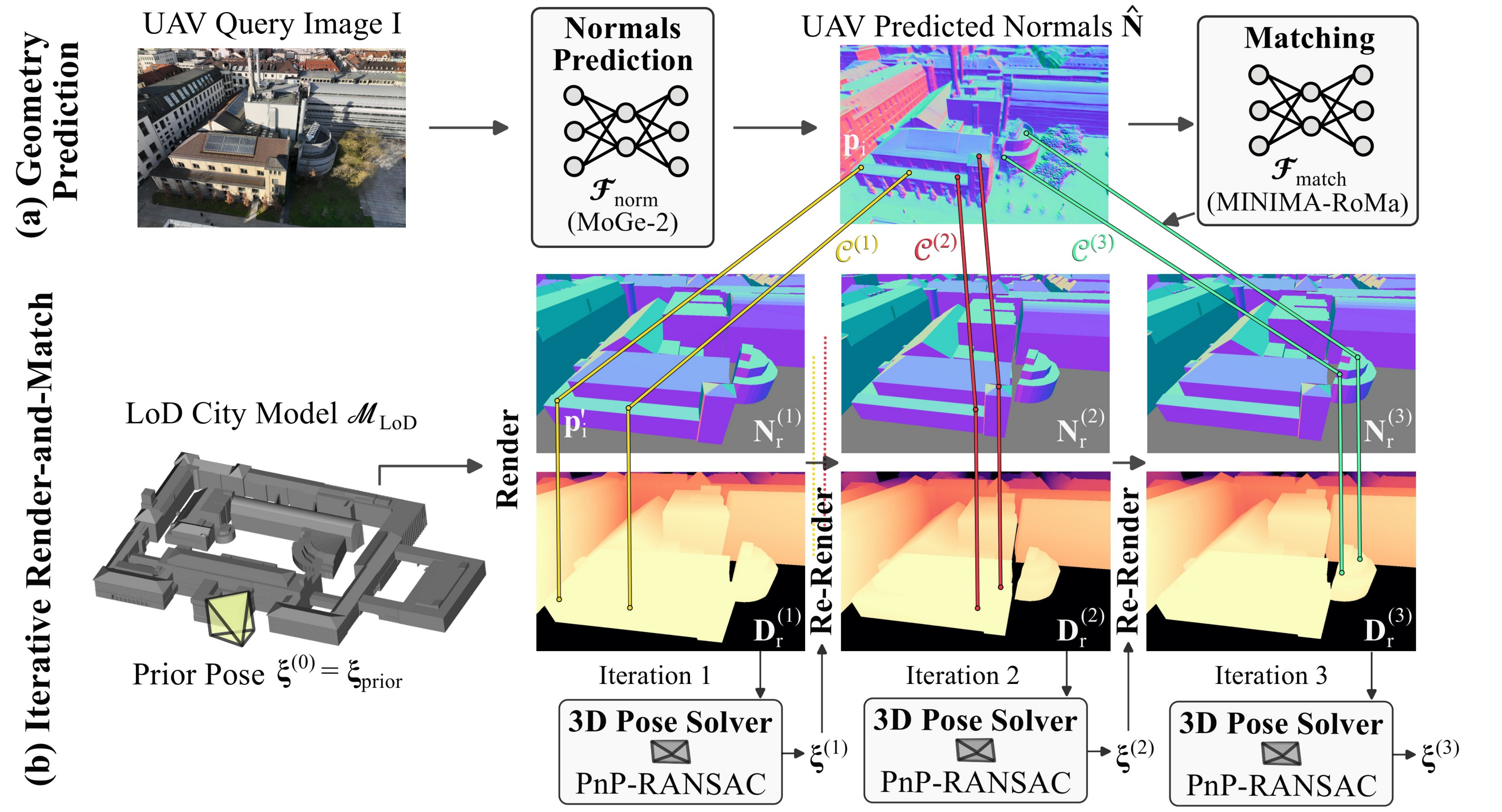
Limit: unreliable plane extraction or association.

Silhouette ambiguity: different poses can produce similar building outlines.
Overlap scores provide no explicit 2D–3D correspondences.

## TL;DR

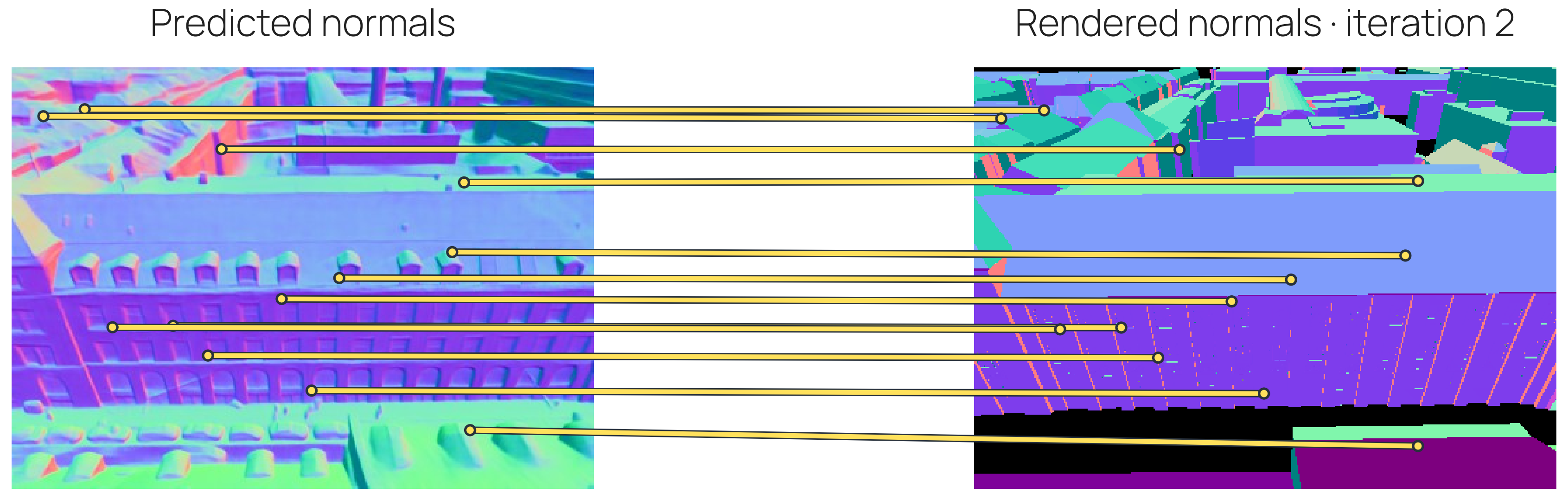
We present MatchCityLoc: surface-normal matching to refine drone poses in textureless city models.
Prior silhouette methods score 2D overlap; we solve pose from 2D–3D matches to reduce ambiguity.

## Render–Match–Refine


(a) Geometry Prediction
UAV Query Image I
Normals Prediction
$\mathcal{F}_{norm}$ (MoGe-2)
UAV Predicted Normals $\hat{N}$
$p_i$
Matching
$\mathcal{F}_{match}$ (MINIMA-RoMa)
$\mathcal{C}^{(1)}$
$\mathcal{C}^{(2)}$
$\mathcal{C}^{(3)}$
(b) Iterative Render-and-Match
LoD City Model $\mathcal{M}_{LoD}$
Prior Pose $\xi^{(0)} = \xi_{prior}$
Render
$p_i'$
$N_r^{(1)}$
$N_r^{(2)}$
$N_r^{(3)}$
$D_r^{(1)}$
$D_r^{(2)}$
$D_r^{(3)}$
Re-Render
Re-Render
Iteration 1
Iteration 2
Iteration 3
3D Pose Solver
PnP-RANSAC
3D Pose Solver
PnP-RANSAC
3D Pose Solver
PnP-RANSAC
$\xi^{(1)}$
$\xi^{(2)}$
$\xi^{(3)}$

1 Predict normals with MoGe-2 [5]. 2 Render LoD normals/depth [8].
3 Match with MINIMA (RoMa) [6, 7], lift to 3D, and solve with PnP.
Repeat with a fixed update budget and validity guards.

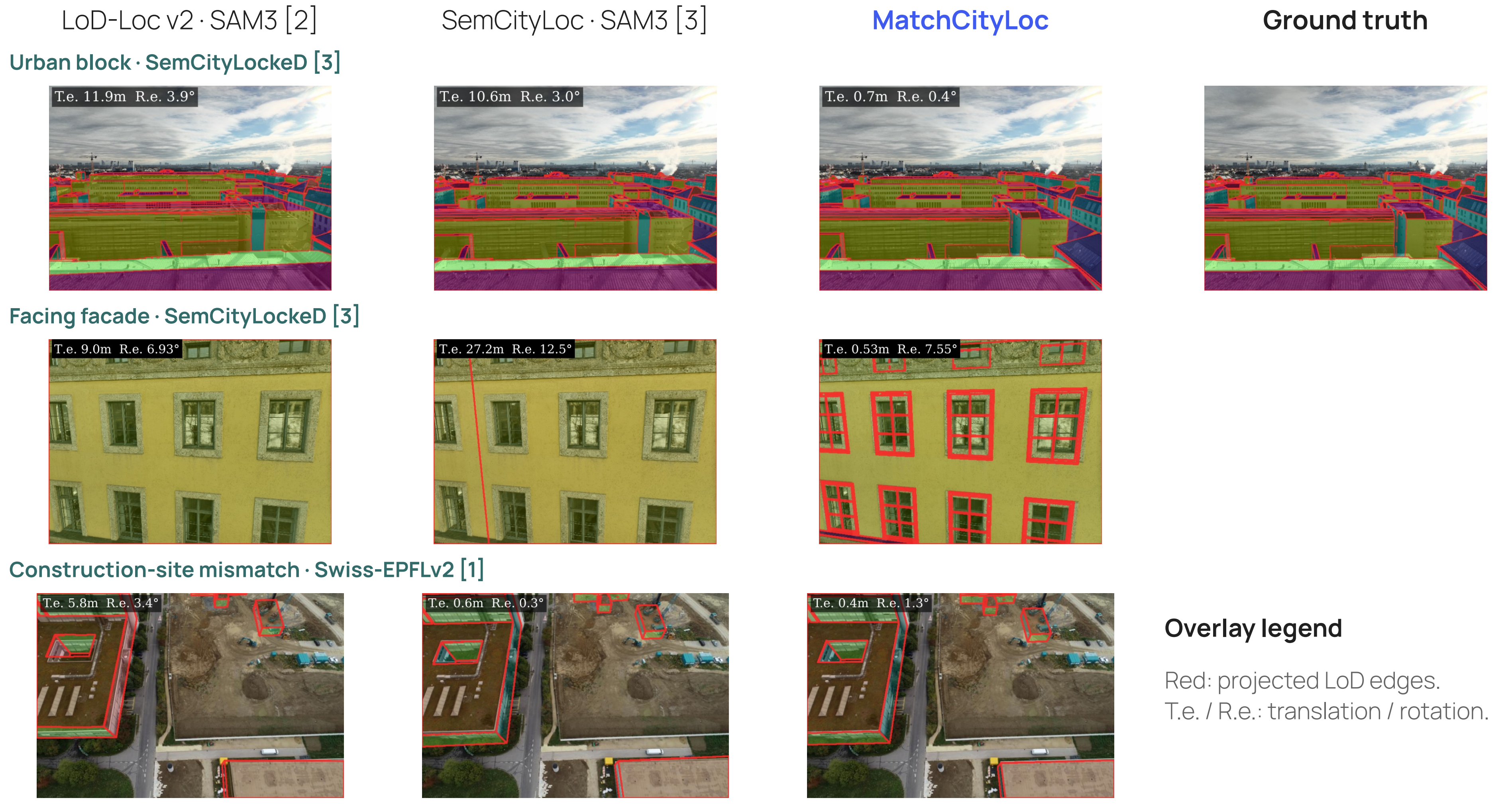
## Normal-to-Normal Matches



Predicted normals · Rendered normals · iteration 2

Matched normal-map pixels are lifted to 3D using rendered depth.

## Qualitative Comparisons


LoD-Loc v2 · SAM3 [2]
SemCityLoc · SAM3 [3]
MatchCityLoc
Ground truth
Urban block · SemCityLockeD [3]
T.e. 11.9m R.e. 3.9°
T.e. 10.6m R.e. 3.0°
T.e. 0.7m R.e. 0.4°
Facing facade · SemCityLockeD [3]
T.e. 9.0m R.e. 6.93°
T.e. 27.2m R.e. 12.5°
T.e. 0.53m R.e. 7.55°
Construction-site mismatch · Swiss-EPFLv2 [1]
T.e. 5.8m R.e. 3.4°
T.e. 0.6m R.e. 0.3°
T.e. 0.4m R.e. 1.3°

**Overlay legend**

Red: projected LoD edges.
T.e. / R.e.: translation / rotation.

## Benchmark Results

**Localization accuracy** — Recall at 2 m / 2° (%) ↑

| | UAVD4L-LoDv2 (LoD3) [1] | | Swiss-EPFLv2 (LoD2) [1] | |
|---|---|---|---|---|
| **Method** | **in-Traj** | **out-Traj** | **in-Place** | **out-Place** |
| LoD-Loc v2 · SAM3† [2] | 35.70 (57.20) | 64.30 (64.90) | 9.40 (74.90) | 17.40 (43.30) |
| SemCityLoc · SAM3† [3] | 63.84 (94.76) | 69.07 (92.20) | 0.00 (70.51) | 0.00 (72.82) |
| PlanarCityLoc [4] | 75.40 | 80.60 | 20.60 | 48.50 |
| **MatchCityLoc (ours)** | **92.64** | **94.66** | **66.50** | **69.30** |

**SemCityLockeD [3]**

| **Method** | **LoD2** | **LoD3** |
|---|---|---|
| LoD-Loc v2 · SAM3† [2] | 37.90 (79.10) | 30.40 (78.00) |
| SemCityLoc · SAM3† [3] | 49.73 (90.07) | 49.63 (93.28) |
| **MatchCityLoc (ours)** | **59.12** | **70.33** |

Off-the-shelf (GT-mask oracle; not a guaranteed upper bound). † Adapted re-runs.
Updates: UAVD4L 3 · SemCityLockeD 5.

## Why Normals?

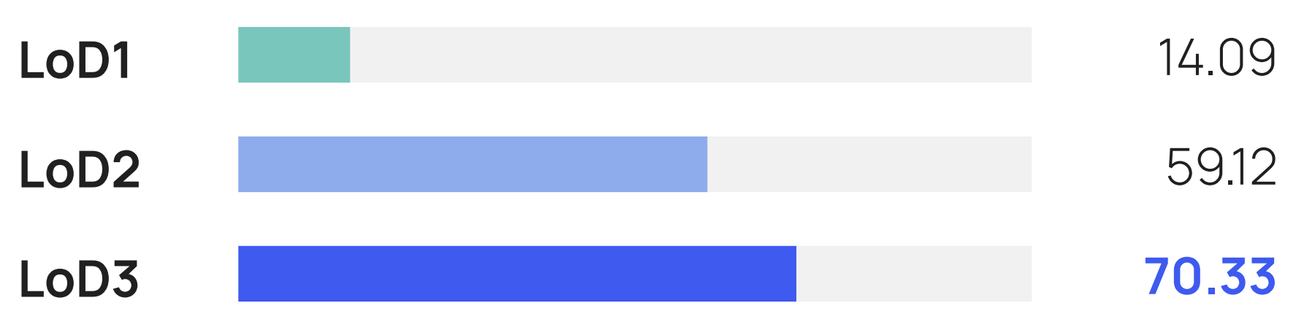
**Recall at 2 m / 2° (%) ↑**

| **Query → model** | **in-Traj** | **out-Traj** |
|---|---|---|
| RGB → Depth | 57.61 | 61.22 |
| RGB → Normals | 77.81 | 83.90 |
| Depth → Normals | 73.38 | 79.24 |
| Depth → Depth | 83.42 | 88.32 |
| **Normals → Normals** | **93.64** | **95.48** |

UAVD4L (LoD3) [1], 3-update runs.

## How Much LoD Detail?

**SemCityLockeD [3] · recall at 2 m / 2° (%) ↑**



Coarse geometry limits normal matching.

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[1] Zhu et al. LoD-Loc: Aerial Visual Localization using LoD 3D Map with Neural Wireframe Alignment. NeurIPS 2024.
[2] Zhu et al. LoD-Loc v2: Aerial Visual Localization over Low Level-of-Detail City Models using Explicit Silhouette Alignment. ICCV 2025.
[3] Mao et al. SemCityLoc: Aerial 6DoF Localization Using Semantic 3D City Models. ECCV 2026.
[4] Dhaouadi et al. PlanarCityLoc: Breaking Silhouette Ambiguity in LoD Aerial Localization via 3D Planes. WACV 2027.
[5] Wang et al. MoGe-2: Accurate Monocular Geometry with Metric Scale and Sharp Details. NeurIPS 2025.
[6] Ren et al. MINIMA: Modality Invariant Image Matching. CVPR 2025.
[7] Edstedt et al. RoMa: Robust Dense Feature Matching. CVPR 2024.
[8] Laine et al. Modular Primitives for High-Performance Differentiable Rendering. ACM TOG 2020.
[9] Peng et al. LoD-Loc v3: Generalized Aerial Localization in Dense Cities using Instance Silhouette Alignment. CVPR 2026.